Chapter 11

"Quantum Tic-Tac-Toe"

Credit – B.C.: "Boomerang" (need permission)

Quantum tic-tac-toe is an example of an abstract quantum system [REF]. If Paradigm is right, that ignoring is the key to abstraction, then quantum tic-tac-toe follows that dictum. Superposition has been limited to two and only two squares, the weights (if you can call them that) are equal but phaseless, there is only one basis (the 9 squares), and no relativity. There is no explicit mathematical operator, but one is implied; two classical marks cannot be in the same square, so the prune by contradiction rule is how entanglement arises. Finally, there is the collapse rule; cyclic entanglements trigger a collapse. In quantum tic-tac-toe, pruning by collapse is done by choice (better play); in physics it is random.

It is the minimum set of quantum features that support an objective measurement process based on self-reference. It remains to be seen if this abstraction is revealing, or merely a clever but low fidelity distortion of quantum mechanics which should be confined to pedagogical purposes.

Distributed Causality

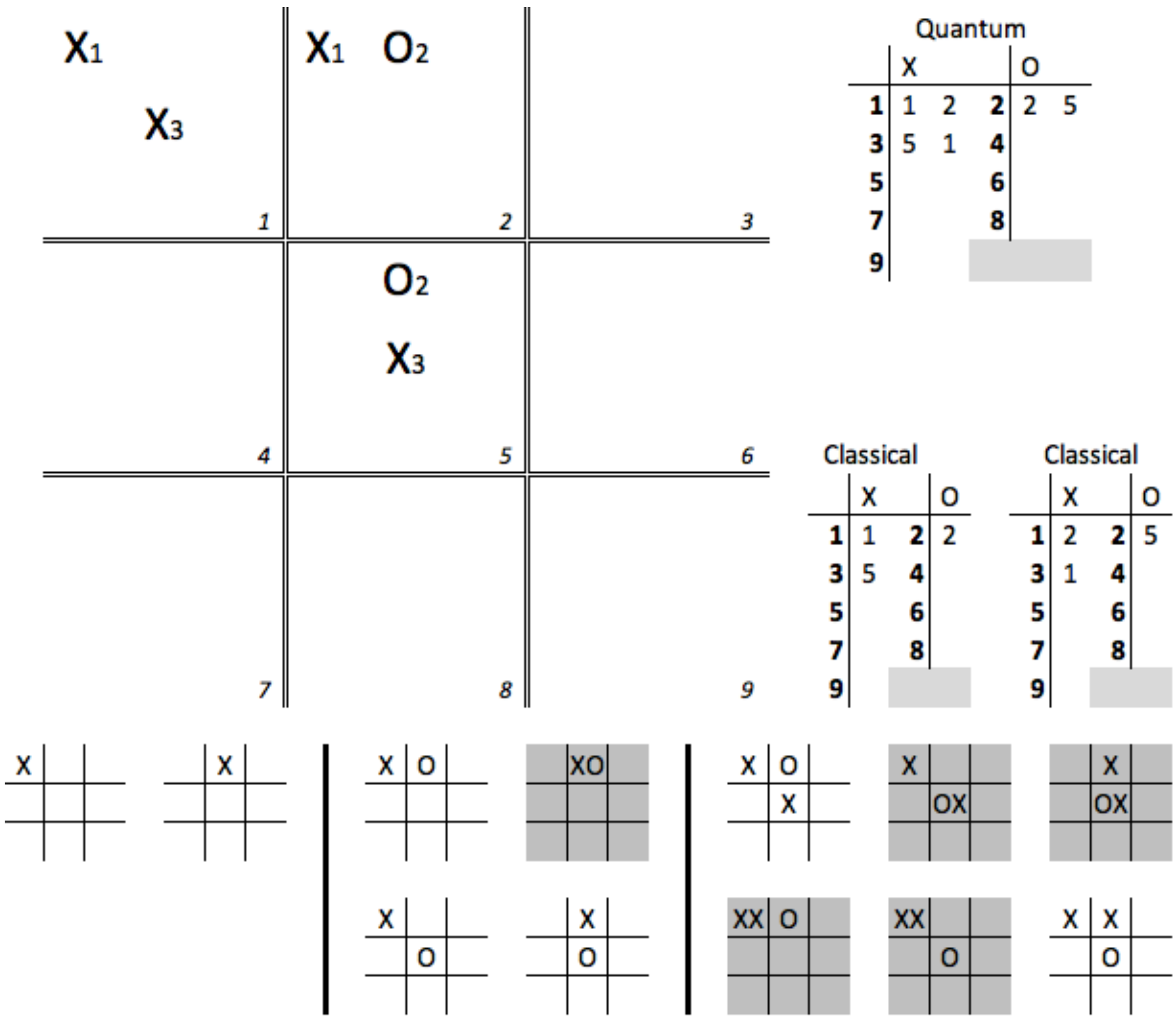


Figure 1 – **Nomenclature for Quantum Tic-Tac-Toe**: The quantum board is indicated by doubled lines, numbered squares, and subscripted *spooky-marks*. A list of quantum moves specifies two squares, one for each spooky mark. A quantum game implies a set of classical games, the *classical ensemble*; three are shown in this figure, the state after each successive move. Contradictory games are pruned from the ensemble (gray background). Upon a cyclic entanglement, a final pruning, a choice, must be made between two possible classical games.

Figure 1 follows Curiosity's lecture. On the upper left is the quantum tic-tac-toe board. Note the doubled lines delineating the squares, how the squares are numbered, and how the spooky-marks are subscripted. Together they make it pretty clear that the quantum rules are in force. To the right of the quantum board is the list of moves in the quantum game; note how two squares must be indicated to specify the locations of the spooky-marks. At the bottom is the classical ensemble in 3 parts; after the 1st move, after the 2nd move, and after the 3rd move but prior to the choice of collapse. Notice that these game boards consist of single lines, unsubscripted marks, and unnumbered squares; implies classical rules in effect. Contradictory games are indicated by being grayed out, which get pruned from existence; they do not progress to the next classical ensemble. Above them and to the right are the two equivalent classic games; the games O must choose between for the pruning by collapse at the end of move 3.

A couple of things should be evident. Cause is now distributed in time; classical time progresses non-linearly, in chunks. The past only becomes fixed when a collapse happens; until then, multiple possible histories are still in play. For the moment, let's be modest, and just call this a metaphor. Within this metaphor, the concepts of time, causality, history, and existence break the current paradigm that guides our usual understanding of reality. In conventional games, each move indicates a tick of the clock. Not in quantum games, and perhaps not in quantum reality either.

Within each *chunk of classical time*…let's pause right there, that is a lousy name for an important concept. Therefore, let's give it a better one, invent some technical jargon; *chronoblock*. Within each chronoblock it appears that the present can influence the past. Yes, but the influencing is strongly constrained. The present cannot change where the spooky-marks were placed; that bit of quantum-ness is fixed. Those original moves allowed for a certain set, a restrained set, of possible histories. That set (the classical ensemble) gets pruned as overlapping superpositions lead to contradictions (can't have more than one classic mark in a square). The final act of pruning is due not to contradictions but to a choice of collapse. In other words, for each chronoblock, possibilities expand, then contract, then collapse at the very last moment; to just one history.

Also, the chronoblocks themselves are not in general linear. Imagine a first isolated move followed by 3 more moves that only entangle moves 2, 3 & 4, with the last of those 3 moves creating a cyclic entanglement. Those moves collapse, leaving the first move still in a superposition. Let move 5 form a simple cyclic entanglement with just the first move. The global history is thus that chronoblock 1, consisting of moves 2, 3 & 4, collapsed at time step 4, but chronoblock 2, consisting of moves 1 & 5, collapsed at time step 5. The second chronoblock spanned the first chronoblock. Note that in this example, we've sequentially labeled the chronoblocks by their collapse move, rather than by their earliest quantum move.

Imagining what this implies for the nature of reality, should this turn out to be more than just metaphor, made Reason's head hurt.

But is this the right viewpoint? Time in the quantum game is still very much linear, no chunking whatsoever; no chronoblocks. Here is the rub. In the lab, when a measurement is made, all that is revealed is a classical state, and by implication a classical history; a history influenced by other realities that now *no longer exist*. They have to be inferred, and in fact, it is this very inferring that led to the development of quantum mechanics. We cannot explain the classical histories without invoking quantum behavior.

That we as a species have been able to figure this out has to count as one of mankind's greatest achievements. It is *so* non-obvious.

Attack the Past

With a little effort it is possible to make this result even more startling. Consider the game in Figure 2. It has been chosen to highlight just how severely the present can influence the past in each chronoblock.

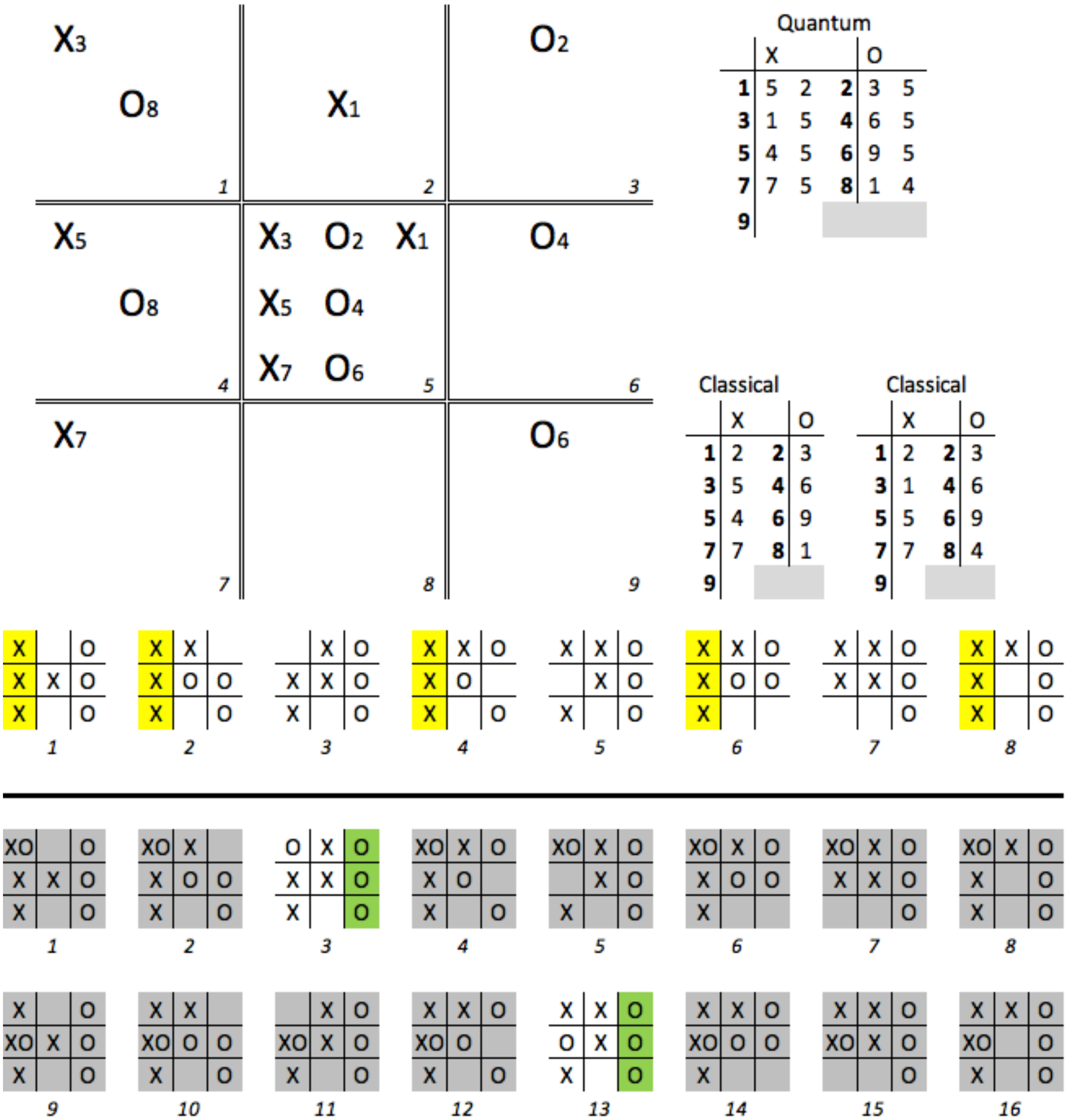


Figure 2 – **Attack the Past**: Two classical ensembles are shown; the first after move 7 and the second after move 8. O's clever move 8 (1-4) has made all five of X's potential wins on move 7 contradictory, pruning them from existence; leaving her

with a win (on move 6!) in both of the only two remaining classical games. Paradox-free time travel.

Up through the first seven moves, all the spooky-marks share the center square. The result is a single large entanglement with only 8 games in the classical ensemble. Note that in 5 of them (games 1, 2, 4, 6, & 8) that player X already has a win in column 1. What's a girl to do? Attack the past. Player O needs a move that will prune by contradiction all 5 of X's winning games, she wants to kill off his successful pasts. She has three such moves, we'll just look at her moving into squares 1 & 4, but the other two options are very similar (4-7 or 1-7).

To see what happens to the classical ensemble on move 8, the ensemble is first duplicated (for a total of 16 games), then in the first row of 8 games an O is placed in square 1 and in the second row of 8 games an O is placed in square 4. Any game that is has two marks in the same square, will be pruned by contradiction. A total of 14 games get pruned, leaving only two games in the ensemble. In neither of them does X have a win. But that's not the best part; in both of them O has a win in the 3rd column, and even that is not the best part. The best part is that his wins didn't occur until move 7 in the chronoblock, but her wins occurred at move 6 of the chronoblock (please convince yourself of this).

Let's review; at time step 7 we detected an outcome we didn't like (X's 5 out of 8 advantage), we took action after that, at time step 8, to make the bad outcome never happen, replacing it at time step 6, *before it never happened*, with an outcome favorable to us. Practical, paradox-free time travel.

Why the emphasis on 'never happened'? Because in the lab, in the real world, all we get from measurement are classical data, the classical moves if you will. Play either classical game listed in Figure 2; X's 5th move is strategically dumb. Given a list of classical moves, it is generally possible to ascertain whether the game was played under classical rules or under quantum rules. This divergence is how scientists were able to determine that we live in a quantum reality, rather than in a classical one.

Now if the idea of time travel makes you nervous, relax. It's not actually time travel, it would be more accurate to describe it as destructive interference in time. Think of it this way. We have entangled the near future with the recent past into an expanded window around now; that's the chronoblock. Within it, causality is dispersed, and the language of time travel is useful, if inaccurate.

This is how Logic kept beating Paradigm; he attacked the past. Trying to talk about causality within a chronoblock makes it sound like time travel is happening, but what is actually going on is that several possibilities (multiple histories), are still possible, are still in play; the past isn't completely fixed yet. At the very end of the chronoblock it is possible to choose which pair of histories (which realities), we find ourselves in. Interestingly, it appears that pruning by contradiction cannot get the number of classical realities in the ensemble to less than two. It's a multivalued proposition. In physics which one gets chosen appears to be random.

This is an inversion of Everett & Wheelers' Many Worlds interpretation [REF]. In their interpretation, the universe splits on every measurement and the variations stay separate and independent. In this version (shall we call it Many Realities?) the splitting occurs upon superposition, while punning occurs upon contradictions and upon measurements. In addition, the resulting classical realties form an integrated whole; they are not separate and independent. There are no copies of you in the other worlds; you, the real complete you, exist *across* all the realities. The potential philosophies about the concept of the self that arise from this metaphor are fascinating.

Table of Objections

Dogma's first objection to spacelike causality is the potential for the Grandfather paradox. It is the strongest argument against time travel, and thus against FTL, for it is an argument from pure logic. Physics, before it is anything else, must be internally consistent; a causal contradiction in time is not.

This is why it is the first entry in our table of objections.

In quantum tic-tac-toe we have an example of a cyclic entanglement in time that is paradox free. Further, it is paradox free in the way anticipated by QTP; it is indeterminant, that is, it is multivalued. Paradigm is gleefully, if prematurely, checking off objection number one.

Half Points

One of the subtleties of quantum tic-tac-toe is a consequence of chronoblocks. Since time advances in chunks it is possible for a collapse to yield a game in which both X and O have each achieved a classical 3-row. However, one of them will have achieved their 3-row earlier in classical time. That player gets a full point, the other one a half point; the justification being that while the losing player couldn't prevent their opponent from winning, they were able to stall the win long enough for them to also eke out a 3-row as well. A real-world example might help; a mother teaching her child how to play, notices the tears well up when the kid has just lost the game. Thinking quickly, she says, "but it's your turn now, and if you move *here*" pointing, "you can also have a 3-row." Kind of like that.

Curiosity alluded to this in the discourse.

Perfect Play

In game theory, perfect play is the idea that one side or the other has a winning strategy. It is not known what perfect play is for chess, but the supposition is that white should be able to force a mate. If not correct, then perfect play is a stalemate as there are proofs that black cannot force a win. In the oriental game of go, perfect play is also unknown. Both of these games have a game tree with an astronomical number of nodes.

In contrast, perfect play for tic-tac-toe is known; it's the cat's game, no winner, just a tie. The reason is that there are only 9! possible games, and an exhaustive analysis of the game tree is not difficult. What was unknown at the time of publication (AJP, 2006) [REF] was whether or not X had a winning strategy in quantum tic-tac-toe.

In 2009 (?) two Japanese grad students (names) from university (name) looking for a project of suitable size to test out some computation optimization ideas, settled on quantum tic-tac-toe as the test vehicle as it had a game tree of the right size and perfect play was unknown. They wrote a computer program to exhaustively search the game tree (about $9!^2$ nodes) and were able to determine that X does in fact have a winning strategy, but O gets a half point [REF]. X can always force a 3-row for himself, but he cannot prevent O from also getting one a bare single move later. O cannot force a cat's game (much less a win), but she can delay the collapse until she also gets a 3-row, just cannot make it happen earlier in time than X's.

This result is behind Reason's observation that once again what is impossible in a classical reality, can sometimes be possible in a quantum version of it.

Interference

Interference is a wave phenomenon; it requires something to oscillate; something with frequency, amplitude, and phase. However, the most counter-intuitive aspect of interference is destructive interference; something prohibited when there are multiple options, but allowed when there is only one. Despite that there are no waves and despite that the 'weights' are phaseless, quantum tic-tac-toe can demonstrate destructive interference.

Superposition alone, the difference between having either of two options but not both, versus the situation of having both, is sufficient to set up situations of destructive interference. This is the situation that Curiosity faced when she attempted to get both to work and get coffee while Reason's leaky uncertainty box was in her car. Both routes open, work *but no* coffee; one or the other route closed, work *and* coffee.

To demonstrate destructive interference in quantum tic-tic-toe takes a little bit of imagination. We have to consider 3 quantum games, the third of which is the superposition of the first two. The setup is shown in Figure 3. In the first two games, a collapse occurred on move three that left all the marks in their classical states (large bold letters). In both of these games, X has an open 2-row in the first column. If these were classical games, O would have no choice (strategically speaking) but to play in the open square; square 7 in the first game, square 4 in the second game. However, this is a quantum game, and she can only place one of her two spooky marks in the open square. She chooses to place the other spooky mark in the third column to strengthen her presence there. Figure 3 shows her moves as 7-9 in game 1 and 4-6 in game 2.

Here is the analysis of why this is her best move. If X makes the same move (7-9 in the 1st game, 4-6 in the 2nd) he creates a cyclic entanglement, which O gets to collapse, and will do so to thwart X's 3-row. It's a failed gambit. If he doesn't, and he has many options, O can simply repeat her move, and regardless of which way X chooses the collapse, she blocks X's open 2-row and creates one for herself. In this case whether her 2nd move ends up in the first column or the third column is random, (X has no strategic preference for either outcome). In other words, in either case, she has a 50/50 chance of her second move ending up in either column.

Now look at game 3; the superposition of games 1 & 2 as of move three. In this game a collapse occurred at move two leaving the first and second moves of the game as classical marks (squares 1 & 3). At this stage, X's 2nd move (X_3) is half in square 4 and half in square 7. He is threatening to repeat this move and regardless of how O collapses it, it will give X a 3-row in the first column by move 5 of the game; unbeatable. To thwart this, she must place both of her spooky marks in column one; one in square 4 the other in square 7. Now no matter how the game resolves she has zero chance that her 2nd move will end up in column three. Destructive interference. What was possible in either of the first two games, becomes impossible in their superposition.

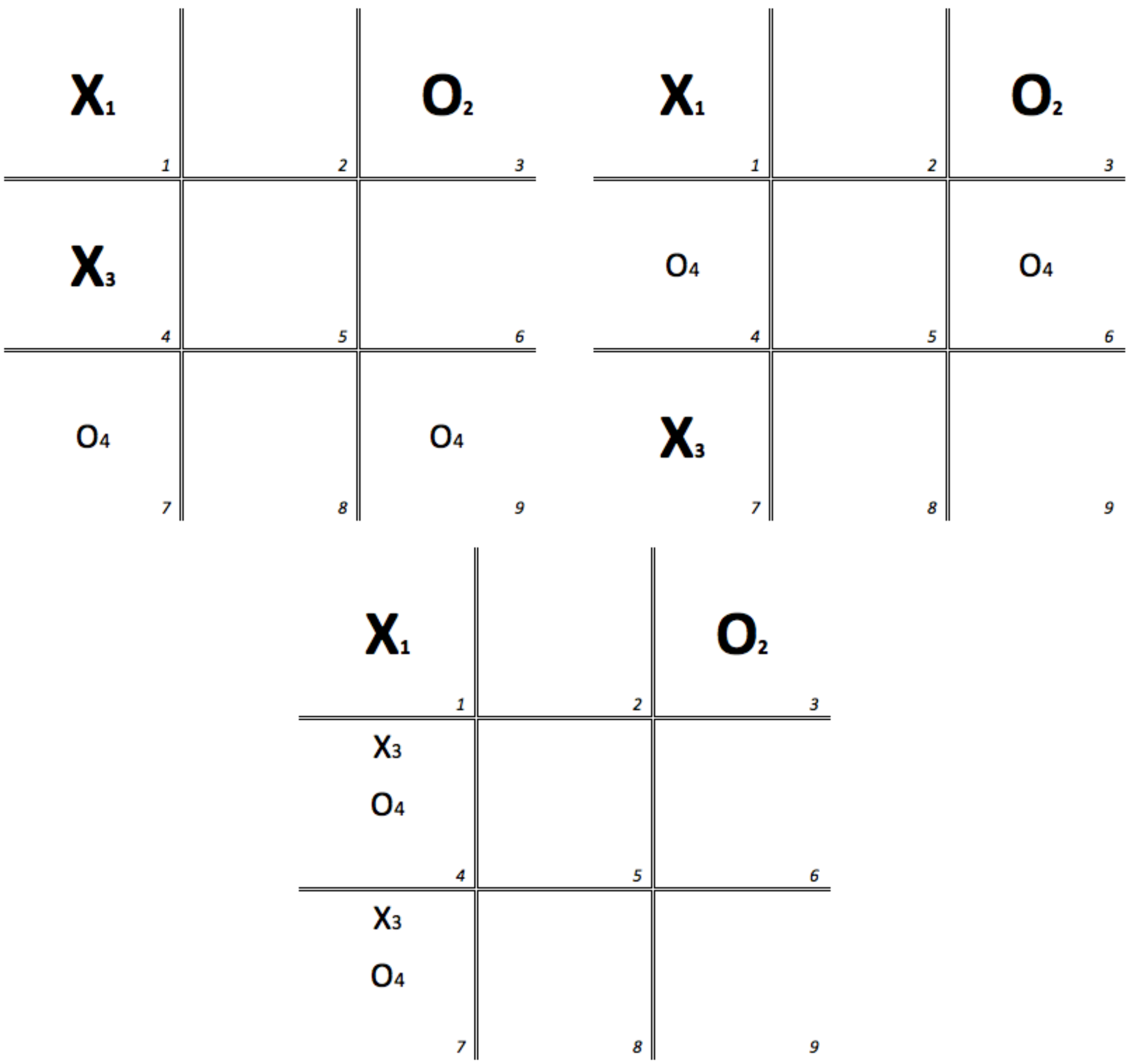


Figure 3 – **Destructive Interference**: As of move 3, the third game is a superposition of the first two games. O's best move in games 1 & 2 give her a 50%

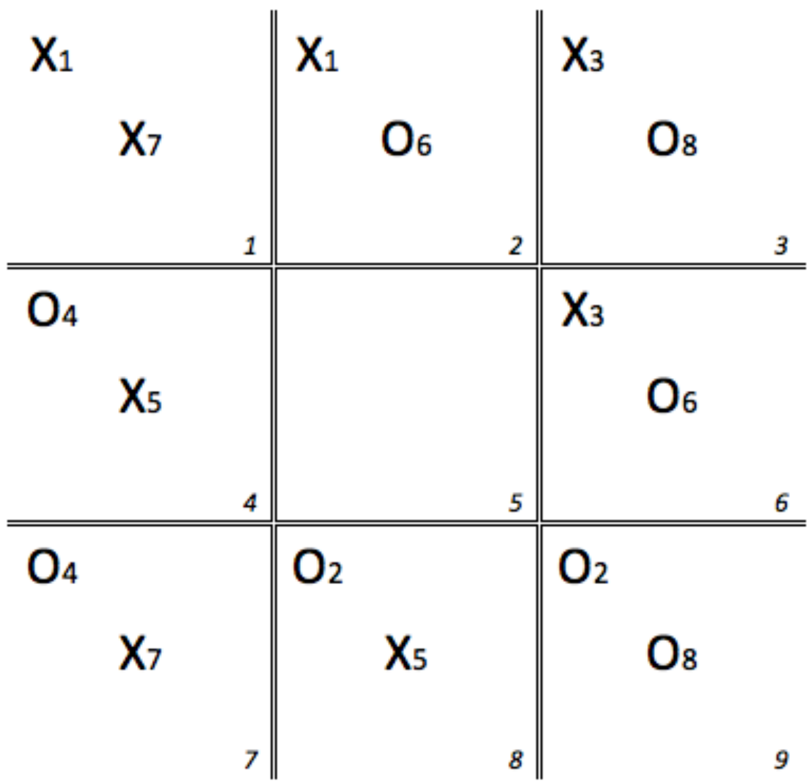
chance of move 4 ending up in column 3, but her best move in game 3, results in no chance at all.

This one still surprises me. Quantum tic-tac-toe may be only a metaphor, but it is a surprisingly rich one. And it's all based on superposition, or if you will, multivalued-ness.

Last Move

The last subtlety that Curiosity hinted at in the Discourse was that while it is not allowed in general to place both spooky marks in the same square, there is one exception; when collapse happens on the 8th move leaving only a single empty square; then this rule can be violated.

Figure 4 shows a single entanglement, a cyclic entanglement, involving all squares except the center. However it collapses, X will be allowed to place his final move in square 5; both halves of his spooky-mark in the same square, which instantly collapses to a classical mark.



Quantum

	X			O	
1	1	2	**2**	9	8
3	3	6	**4**	7	4
5	4	8	**6**	2	6
7	1	7	**8**	9	3
9					

Classical

	X		O
1	1	**2**	9
3	6	**4**	4
5	8	**6**	2
7	7	**8**	3
9			

Classical

	X		O
1	2	**2**	8
3	3	**4**	7
5	4	**6**	6
7	1	**8**	9
9			

X	O	O
O		X
X	X	O

X	X	X
X		O
O	O	O

Figure 4 – **Last Move Exception**: If there is only one square where moves can be made, both spooky marks of the last move may be placed there.

A rather trivial point. But your pernicious author has saved it for last, for this cyclic entanglement around the perimeter of the quantum tic-tac-toe board has interesting properties.

This overcomes the self-reference objection.